

Mood-Based Music Recommendation System

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Objective:

To build an intelligent music recommendation system that detects the user's mood through facial expression in real time and recommends songs accordingly by searching on YouTube.

Motivation:

Music has a deep connection to emotions. Automating music recommendations based on facial expressions makes the user experience more intuitive and personal. This project helps users find the perfect song for their mood without manual searching.

How It Works:

1. Data Collection (dc folder):

Python scripts like happy.py, sad.py, etc., collect facial expression data using Mediapipe. Data is saved for model training.

2. Data Training (dt folder):

The dt.py script trains a machine learning model using TensorFlow based on the collected data.

3. Inference (inf folder):

Running inf.py opens the webcam and performs real-time facial expression recognition. Based on the detected mood, the system proceeds to recommend music.

Application Workflow:

1. Run inf.py to open webcam and detect the user's facial expression in real time.
2. Select Music Language (e.g., Hindi, English) and Preferred Artist (e.g., Arijit Singh) via web interface (Streamlit).
3. Based on detected mood, a YouTube search is performed, such as: 'Hindi Sad Arijit Singh music'.

Technologies Used:

- Mediapipe: Real-time face detection and landmark extraction.
- TensorFlow: Model training and expression classification.

- Streamlit & Streamlit-wrt: Interactive web interface.
- YouTube Search API: For music recommendations.

Installation:

`pip install mediapipe tensorflow streamlit streamlit-wrt`

Advantages:

- Real-time and automated mood detection.
- Simple and intuitive web interface.
- Personalized music recommendations.
- Open-source and customizable.

Future Improvements:

- Expand dataset for improved accuracy.
- Add more moods (e.g., excited, bored).
- Improve recommendation logic using user feedback.

Conclusion:

This system effectively integrates data collection, model training, and real-time inference to offer personalized music recommendations based on facial expression. It enhances user experience and shows the power of machine learning combined with computer vision.